

Integration — Lesson 26 Test: u-Substitution Mastery

10 questions · show your work in the box · no calculator · exact values only — leave π , e , $\ln$ in your answers · on definite integrals with u-sub, show your changed limits

Name: _____ Date: _____ Score: _____ / 10

1. Compute $\int_0^1 2x(x^2+2)^2 dx$ by u-substitution with changed limits (do not back-substitute). 1 pt

2. Just find the new u-limits (no integrating): (a) $u = x^3+1$ with x from 0 to 2 (b) $u = \cos x$ with x from 0 to $\pi/2$. 1 pt

3. Find $\int x/(x^2+9) dx$. 1 pt

4. Compute $\int_1^{e^2} (\ln x)/x \, dx$ with changed limits.

1 pt

5. Find $\int \cos^4 x \sin x \, dx$. (Careful — a minus sign is hiding in this one.)

1 pt

6. Compute $\int_0^{\pi/2} \sin^4 x \cos x \, dx$ with changed limits.

1 pt

7. Find $\int e^x/(e^x+5) \, dx$, and name the general pattern you used.

1 pt

8. Compute $\int_0^{\pi/4} \sec^2 x / (1 + \tan x) \, dx$ exactly. (Hint: what is the derivative of the bottom?)

1 pt

9. A student computes $\int_0^1 x(x^2+1)^2 \, dx$: " $u = x^2+1$, $x \, dx = du/2$, so the integral is $(1/2) \int_0^1 u^2 \, du = (1/2)(1/3) = 1/6$." Find the error and give the correct answer.

10. Compute $\int_0^{\pi/4} \tan^3 x \sec^2 x \, dx$ with changed limits. (Recall $(\tan x)' = \sec^2 x$.)

1 pt
